

Milestone 4: Equity, Forecasting & Finalization

Objective: Evaluate mobility accessibility across the city, forecast future demand, and deliver executive-level intelligence to support data-driven urban mobility decisions.



Milestone 4 Scope

- Mobility Access Index & Equity
- Demand Forecasting & Anomalies
- Executive Intelligence Dashboard
- Performance & Deployment Readiness

Mobility Access & Equity

Mobility Access Index (MAI)

A composite 0–100 score combining transit frequency, stop density, travel-time burden and affordability to measure how easily residents in each zone can reach opportunity.



Accessibility by Zone

MAI computed per geographic zone to localize gaps precisely.



Demographic Comparison

Access scores cross-referenced with income, age and vehicle-ownership data.



Underserved Areas

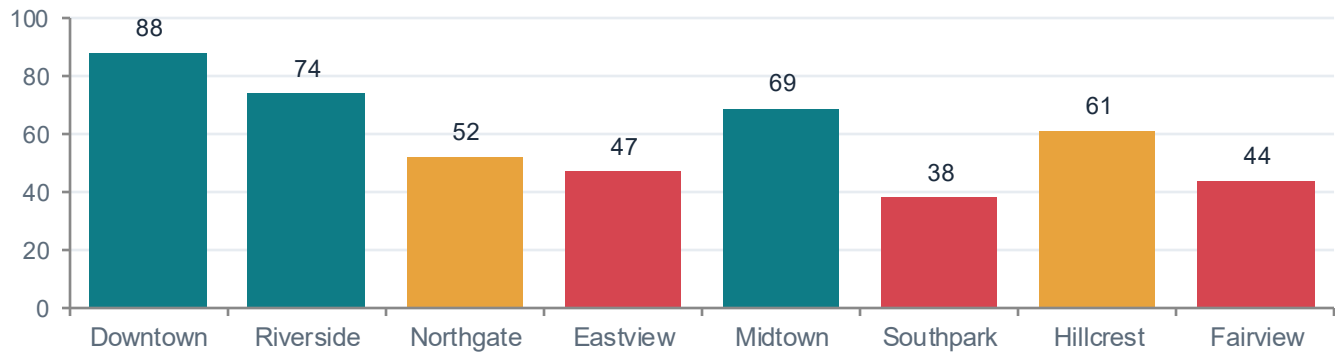
Zones below the equity threshold flagged for targeted intervention.



Mobility Gaps

Service gaps quantified to prioritize routes, stops and frequency upgrades.

Mobility Access Index by Zone



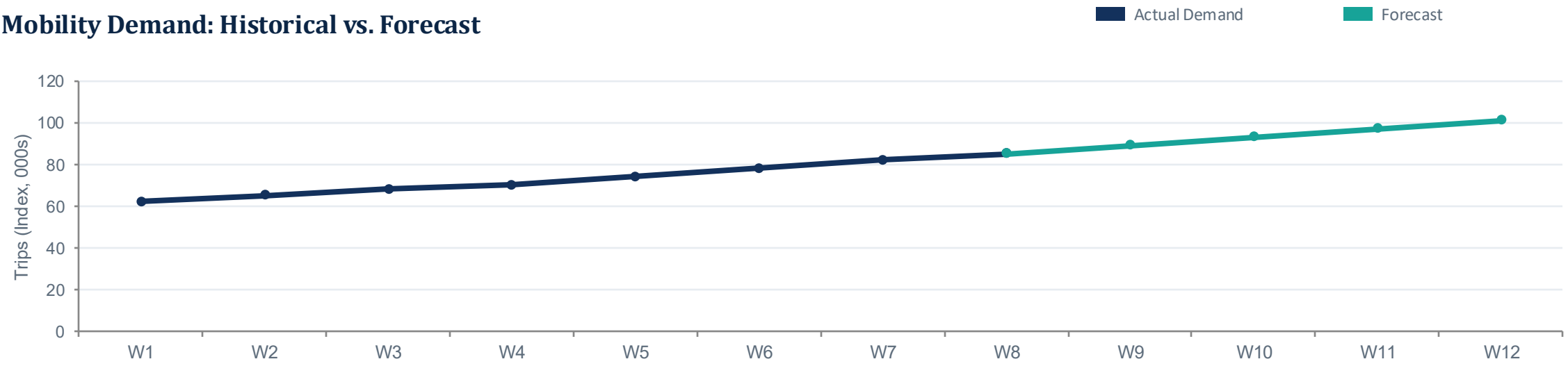
Zone Accessibility Map



High Access (MAI ≥ 65) Moderate (50–64) Underserved (< 50)

Forecasting & Anomaly Detection

Mobility Demand: Historical vs. Forecast



Trend & Deviation Drivers

-  Steady +5.2% week-over-week growth trend across the forecast horizon.
-  Weather events (rain, heat advisories) correlate with short-term demand dips.
-  City events and holidays produce demand spikes of 15–30% above baseline.

Detected Demand Anomalies

Date	Zone	Deviation	Likely Cause
Mar 12	Downtown	+27%	Marathon event
Mar 19	Riverside	-18%	Severe storm
Apr 02	Midtown	+21%	Concert / venue event
Apr 15	Southpark	-14%	Transit line closure

Executive Intelligence Dashboard



Final Platform & Outcomes

